An Intercalation Calculus with Partial Proof Terms

Ana Catarina Sousa
LIACC - Artificial Intelligence and Computer Science Laboratory

Joint work with José Espírito Santo

Women in Logic Workshop
Lisbon, Portugal

24th July 2026

Fundação para a Ciência e a Tecnologia

Proof Search

Proof Search

- Looking for proofs/derivations;

Proof Search

- Looking for proofs/derivations;
- Theory based on Sequent Calculi:

Proof Search

- Looking for proofs/derivations;
- Theory based on Sequent Calculi:
    - Gentzen Sequent Calculus;
    - G3 systems;
    - Focusing;
    - ...

Proof Search

- Looking for proofs/derivations;
- Theory based on Sequent Calculi:
    - Gentzen Sequent Calculus;
    - G3 systems;
    - Focusing;
    - ...
- Theorem Provers:

Proof Search

- Looking for proofs/derivations;
- Theory based on Sequent Calculi:
    - Gentzen Sequent Calculus;
    - G3 systems;
    - Focusing;
    - ...
- Theorem Provers:
    - Resolution;
    - Tableaux.

Proof Search

- Looking for proofs/derivations;
- Theory based on Sequent Calculi:
  - Gentzen Sequent Calculus;
  - G3 systems;
  - Focusing;
  - ...
- Theorem Provers:
  - Resolution;
  - Tableaux.

Problem

Completeness and Efficiency	✓
Human Mathematical Reasoning	✗

Proof Search

- Looking for proofs/derivations;
- Theory based on Sequent Calculi:
  - Gentzen Sequent Calculus;
  - G3 systems;
  - Focusing;
  - ...
- Theorem Provers:
  - Resolution;
  - Tableaux.

Alternative

Proof Search in **Natural Deduction**

Why Natural Deduction?

Why Natural Deduction?

*"…(**Natural Deduction** exhibits) a **close affinity to actual reasoning**, which had been our fundamental aim in setting up the calculus. The calculus lends itself in particular to the **formalization of mathematical proofs**."*

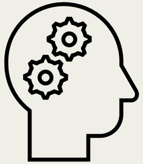
Gerhard Gentzen

Proof Search in Natural Deduction

Improve the **modeling of mathematical reasoning** through **Natural Deduction**.

Proof Search in Natural Deduction

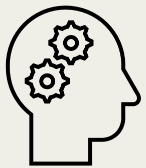
Improve the **modeling of mathematical reasoning** through **Natural Deduction**.



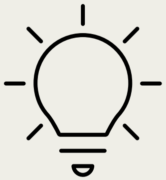
Reasoning is
Proof Search

Proof Search in Natural Deduction

Improve the **modeling of mathematical reasoning** through **Natural Deduction**.



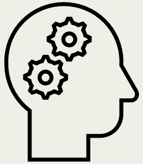
Reasoning is
Proof Search



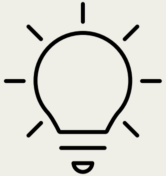
Proof Search
obtains proofs in
Natural Deduction

Proof Search in Natural Deduction

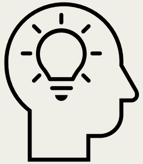
Improve the **modeling of mathematical reasoning** through **Natural Deduction**.



Reasoning is
Proof Search



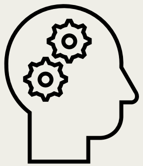
Proof Search
obtains proofs in
Natural Deduction

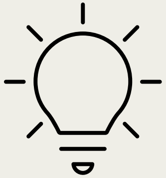


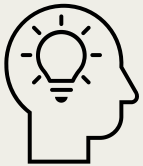
Proof Search
is made directly in
Natural Deduction

Proof Search in Natural Deduction

Improve the **modeling of mathematical reasoning** through **Natural Deduction**

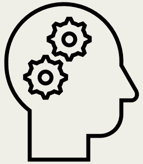


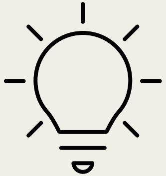


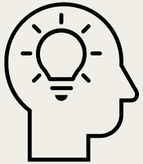


Proof Search in Natural Deduction

Improve the **modeling of mathematical reasoning** through **Natural Deduction**







Partial Proof Terms

Partial Proofs & Partial Proof Terms

Finished Proof

$$\dfrac{\dfrac{\overline{x : A, y : B \vdash A}\ Ax}{x : A \vdash B \supset A}\ I\supset}{\vdash A \supset B \supset A}\ I\supset$$

Finished Proof

$$\dfrac{\dfrac{\overline{x : A, y : B \vdash A}\ Ax}{x : A \vdash B \supset A}\ I \supset}{\vdash A \supset B \supset A}\ I \supset$$

Curry
Howard
Isomorphism

Finished Proof

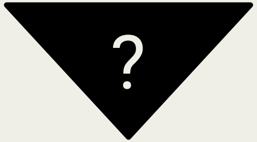
Proof Term

$$\dfrac{\dfrac{\overline{x : A, y : B \vdash A}\ Ax}{x : A \vdash B \supset A}\ I\supset}{\vdash A \supset B \supset A}\ I\supset$$

$\xrightarrow{\text{Curry Howard Isomorphism}}$

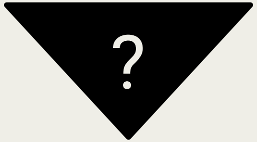
$$\lambda x^A.\lambda y^B.x$$

Unfinished Proof



$$\frac{x : A \vdash B \supset A}{\vdash A \supset B \supset A}\ I\supset$$

Unfinished Proof

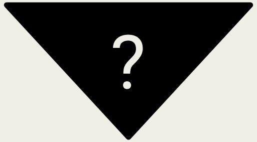


$$\dfrac{x : A \vdash B \supset A}{\vdash A \supset B \supset A}\ I \supset$$

Partial Proof Term

$$\lambda x^{A}.(\underline{x : A \vdash B \supset A})$$

Unfinished Proof



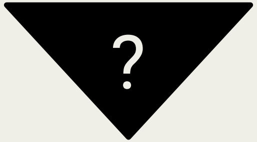
$$\frac{x : A \vdash B \supset A}{\vdash A \supset B \supset A}\ I\supset$$

Partial Proof Term

$$\lambda x^{A}.(\underline{x : A \vdash B \supset A})$$

- **Formal Sequents**

Unfinished Proof



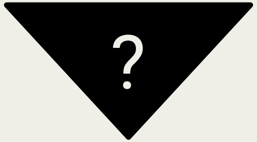
$$\frac{x : A \vdash B \supset A}{\vdash A \supset B \supset A}\; I \supset$$

Partial Proof Term

$$\lambda x^{A}.(\underline{x : A \vdash B \supset A})$$

- Formal Sequents
- **Calculus of Partial Derivations**

Unfinished Proof



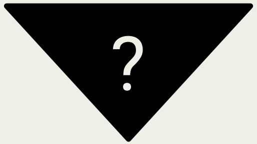
$$\dfrac{x : A \vdash B \supset A}{\vdash A \supset B \supset A}\; I \supset$$

Partial Proof Term

$$\lambda x^{A}.(\underline{x : A \vdash B \supset A})$$

- Formal Sequents
- Calculus of Partial Derivations
- **Proof Search as Rewriting**

Unfinished Proof



$$\dfrac{x : A \vdash B \supset A}{\vdash A \supset B \supset A}\ I\supset$$

Partial Proof Term

$$\lambda x^{A}.(\underline{x : A \vdash B \supset A})$$

- Formal Sequents
- Calculus of Partial Derivations
- Proof Search as Rewriting
- **Idealized Proof Search**

A Focused Intercalation in Natural Deduction

NJT

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B}\ I \qquad \frac{\Gamma \rhd p}{\Gamma \vdash p}\ C$$

$$\frac{}{\Gamma, x : A \rhd A}\ A \qquad \frac{\Gamma \rhd A \supset B \quad \Gamma \vdash A}{\Gamma \rhd B}\ E$$

NJT with Proof Terms

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \lambda x^A . M : A \supset B}\ I \qquad \frac{\Gamma \rhd H : p}{\Gamma \vdash app(H) : p}\ C$$

$$\frac{}{\Gamma, x : A \rhd x : A}\ A \qquad \frac{\Gamma \rhd H : A \supset B \quad \Gamma \vdash N : A}{\Gamma \rhd HN : B}\ E$$

NJT with Proof Terms

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \underline{\lambda x^A . M} : A \supset B}\ I \qquad \frac{\Gamma \rhd H : p}{\Gamma \vdash \underline{app(H)} : p}\ C$$

$$\frac{}{\Gamma, x : A \rhd x : A}\ A \qquad \frac{\Gamma \rhd H : A \supset B \quad \Gamma \vdash N : A}{\Gamma \rhd HN : B}\ E$$

Proof Terms

NJT with Proof Terms

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \lambda x^A.M : A \supset B}\ I \qquad \frac{\Gamma \rhd H : p}{\Gamma \vdash app(H) : p}\ C$$

$$\frac{}{\Gamma, x : A \rhd \underline{x} : A}\ A \qquad \frac{\Gamma \rhd H : A \supset B \quad \Gamma \vdash N : A}{\Gamma \rhd \underline{HN} : B}\ E$$

Neutral Terms

NJT (complete) derivation

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{f : p \supset p, x : p \rhd f : p \supset p}\,A \qquad \dfrac{\dfrac{}{f : p \supset p, x : p \rhd x : p}\,A}{f : p \supset p, x : p \vdash \mathsf{app}(x) : p}\,C}{f : p \supset p, x : p \rhd f\mathsf{app}(x) : p}\,E}{f : p \supset p, x : p \vdash \mathsf{app}(f\mathsf{app}(x)) : p}\,C}{f : p \supset p \vdash \lambda x^{p}.\mathsf{app}(f\mathsf{app}(x)) : p \supset p}\,I}{\vdash \lambda f^{p \supset p}.\lambda x^{p}.\mathsf{app}(f\mathsf{app}(x)) : (p \supset p) \supset p \supset p}\,I
$$

NJT (complete) derivation

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{f : p \supset p, x : p \rhd f : p \supset p}\,A \qquad \dfrac{\dfrac{}{f : p \supset p, x : p \rhd x : p}\,A}{f : p \supset p, x : p \vdash \mathsf{app}(x) : p}\,C}{f : p \supset p, x : p \rhd f\mathsf{app}(x) : p}\,E}{f : p \supset p, x : p \vdash \mathsf{app}(f\mathsf{app}(x)) : p}\,C}{f : p \supset p \vdash \lambda x^{p}.\mathsf{app}(f\mathsf{app}(x)) : p \supset p}\,I}{\vdash \underline{\lambda f^{p \supset p}.\lambda x^{p}.\mathsf{app}(f\mathsf{app}(x))} : (p \supset p) \supset p \supset p}\,I
$$

Focused Intercalation

$$\vdash (p \supset p) \supset p \supset p$$

Focused Intercalation

Bottom-up Inversion Phase

Invert I

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\, I}{\vdash (p \supset p) \supset p \supset p}\, I$$

Focused Intercalation

Bottom-up
Inversion Phase

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\, I}{\vdash (p \supset p) \supset p \supset p}\, I$$

Top-Down
Focusing Phase

Focus Down

Focused Intercalation

$$\dfrac{}{f : p \supset p, x : p \rhd p \supset p}\ A$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\ I}{\vdash (p \supset p) \supset p \supset p}\ I$$

Focused Intercalation

Top-Down
Focusing Phase

$$\frac{}{f : p \supset p, x : p \rhd p \supset p}\ A$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\ I}{\vdash (p \supset p) \supset p \supset p}\ I$$

Focused Intercalation

Top-Down
Focusing Phase

Keep the focus

$$\dfrac{\dfrac{}{f : p \supset p, x : p \rhd p \supset p}\,A \qquad f : p \supset p, x : p \vdash p}{f : p \supset p, x : p \rhd p}\,E$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I$$

Focused Intercalation

Top-Down Focusing Phase

Keep the focus

?

$$\dfrac{\dfrac{}{f : p \supset p, x : p \rhd p \supset p}\,A \qquad f : p \supset p, x : p \vdash p}{f : p \supset p, x : p \rhd p}\,E$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I$$

Top-Down
Focusing Phase

Focused Intercalation

?

$$\dfrac{\dfrac{}{f : p \supset p, x : p \triangleright p \supset p}\,A \qquad f : p \supset p, x : p \vdash p}{f : p \supset p, x : p \triangleright p}\,E$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I$$

Focused Intercalation

Top-Down
Focusing Phase

Finish the focus

?

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{f : p \supset p, x : p \rhd p \supset p}\,A \qquad f : p \supset p, x : p \vdash p}{f : p \supset p, x : p \rhd p}\,E}{f : p \supset p, x : p \vdash p}\,C}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I
$$

Focused Intercalation

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{f : p \supset p, x : p \rhd p \supset p}\,A \qquad \dfrac{\dfrac{}{f : p \supset p, x : p \rhd p}\,A}{f : p \supset p, x : p \vdash p}\,C}{f : p \supset p, x : p \rhd p}\,E}{f : p \supset p, x : p \vdash p}\,C}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I$$

Focused Intercalation

$$
\dfrac{
\dfrac{
\dfrac{}{f : p \supset p, x : p \rhd \dfrac{}{}\!\!\!\;\; f : p \supset p}\;A
\qquad
\dfrac{\dfrac{}{f : p \supset p, x : p \rhd x : p}\;A}{f : p \supset p, x : p \vdash \mathsf{app}(x) : p}\;C
}{
\dfrac{f : p \supset p, x : p \rhd f\mathsf{app}(x) : p}{f : p \supset p, x : p \vdash \mathsf{app}(f\mathsf{app}(x)) : p}\;C
}\;E
}{
\dfrac{f : p \supset p \vdash \lambda x^{p}.\mathsf{app}(f\mathsf{app}(x)) : p \supset p}{\vdash \underline{\lambda f^{p \supset p}.\lambda x^{p}.\mathsf{app}(f\mathsf{app}(x))} : (p \supset p) \supset p \supset p}\;I
}\;I
$$

Questions

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash {\color{olive}?} : p}{f : p \supset p \vdash {\color{olive}?} : p \supset p}\; I}{\vdash {\color{olive}?} : (p \supset p) \supset p \supset p}\; I$$

Questions

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash ? : p}{f : p \supset p \vdash ? : p \supset p}\, I}{\vdash ? : (p \supset p) \supset p \supset p}\, I$$

1. How to annotate partial derivations?

Questions

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash {?} : p}{f : p \supset p \vdash {?} : p \supset p}\,I}{\vdash {?} : (p \supset p) \supset p \supset p}\,I$$

1. How to annotate partial derivations?
2. **How to formalize proof search?**

FROM NJT TO $\mathcal{N}JT_\partial$

FROM *NJT* TO $\mathcal{N}JT_{\partial}$

Proof Terms

$$\begin{aligned} M, N &::= \lambda x^A.M \,|\, \mathsf{app}(H) \\ H &::= x \,|\, H N \end{aligned}$$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Partial Proof Terms

$$
\begin{aligned}
M, N &::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p) \\
H &::= x \mid H N
\end{aligned}
$$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p)$$
$$H ::= x \mid H N$$

Formal Sequent

$\underline{\Gamma \vdash A}$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p)$$
$$H ::= x \mid H N$$

Formal Sequents

$$\underline{\Gamma \vdash A}$$
$$\underline{\Gamma \rhd A}$$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p)$$
$$H ::= x \mid H\,N$$

Partial Sequents

$$\Xi \Vdash M : \sigma$$
$$\Xi \Vdash H : \rho$$

FROM *NJT* TO $\mathcal{NJT}_\partial$

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p)$$
$$H ::= x \mid H N$$

Partial Sequents

$$\boxed{\Xi \Vdash M : \sigma}$$ **Proof State**

$$\Xi \Vdash H : \rho$$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p)$$

$$H ::= x \mid H N$$

Partial Sequents

$$\boxed{\Xi \Vdash M : \sigma}$$

Proof State

$$\Xi \Vdash H : \rho$$

Goal Sequent

Record of Search

Proof Obligations

FROM *NJT* TO $\mathcal{N}JT_{\partial}$

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid \mathsf{app}(H) \mid \underline{\sigma} \mid \mathsf{app}(H, \underline{\rho}, p)$$

$$H ::= x \mid H N$$

Partial Sequents

$$\boxed{\Xi \Vdash M : \sigma}$$ **Proof State**

$$\Xi \Vdash H : \rho$$ **Goal Sequent**

Record of Search

Proof Obligations

$\underline{\sigma}$ $\quad (\underline{\rho}, p)$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Inference Rules

$$\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Xi \Vdash H : (\Gamma \rhd p)}{\Xi \Vdash \mathsf{app}(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\epsilon \Vdash x : (\Gamma, x : A \rhd A)}\ A \qquad \frac{\Xi_1 \Vdash H : (\Gamma \rhd A \supset B) \quad \Xi_2 \Vdash N : (\Gamma \vdash A)}{\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \rhd B)}\ E$$

FROM NJT TO $\mathcal{N}JT_\partial$

Inference Rules

$$\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A . M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Xi \Vdash H : (\Gamma \rhd p)}{\Xi \Vdash \mathsf{app}(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\epsilon \Vdash x : (\Gamma, x : A \rhd A)}\ A \qquad \frac{\Xi_1 \Vdash H : (\Gamma \rhd A \supset B) \quad \Xi_2 \Vdash N : (\Gamma \vdash A)}{\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \rhd B)}\ E$$

FROM *NJT* TO $\mathcal{N}JT_\partial$

Inference Rules

$$\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Xi \Vdash H : (\Gamma \rhd p)}{\Xi \Vdash \mathsf{app}(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\epsilon \Vdash x : (\Gamma, x : A \rhd A)}\ A \qquad \frac{\Xi_1 \Vdash H : (\Gamma \rhd A \supset B) \quad \Xi_2 \Vdash N : (\Gamma \vdash A)}{\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \rhd B)}\ E$$

FROM *NJT* TO $\mathcal{NJT}_\partial$

Inference Rules

$$\frac{}{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma}\ \partial$$

$$\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Xi \Vdash H : (\Gamma \rhd p)}{\Xi \Vdash \mathsf{app}(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\epsilon \Vdash x : (\Gamma, x : A \rhd A)}\ A \qquad \frac{\Xi_1 \Vdash H : (\Gamma \rhd A \supset B) \quad \Xi_2 \Vdash N : (\Gamma \vdash A)}{\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \rhd B)}\ E$$

FROM *NJT* TO $\mathcal{NJT}_\partial$

Inference Rules

$$\frac{}{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma}\ \partial \qquad \frac{\Xi \Vdash H : (\Gamma \rhd A)}{\Xi@[(\underline{\Gamma \rhd A}, p)] \Vdash \mathsf{app}(H, \underline{\Gamma \rhd A}, p) : (\Gamma \vdash p)}\ \partial$$

$$\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Xi \Vdash H : (\Gamma \rhd p)}{\Xi \Vdash \mathsf{app}(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\epsilon \Vdash x : (\Gamma, x : A \rhd A)}\ A \qquad \frac{\Xi_1 \Vdash H : (\Gamma \rhd A \supset B) \quad \Xi_2 \Vdash N : (\Gamma \vdash A)}{\Xi_1@\Xi_2 \Vdash HN : (\Gamma \rhd B)}\ E$$

NJT_∂: Reduction Rules

$$(IR) \qquad \underline{\Gamma \vdash A \supset B} \to \lambda x^A.(\underline{\Gamma, x : A \vdash B})$$

$$(SFD) \qquad \underline{\Gamma, x : A \vdash p} \to \mathsf{app}(x, \underline{\Gamma, x : A \rhd A}, p)$$

$$(KFD) \qquad \mathsf{app}(H, \underline{\Gamma \rhd A \supset B}, p) \to \mathsf{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \rhd B}, p)$$

$$(FFD) \qquad \mathsf{app}(H, \underline{\Gamma \rhd p}, p) \to \mathsf{app}(H)$$

NJT_∂: Reduction Rules

$$(IR) \quad \underline{\Gamma \vdash A \supset B} \to \lambda x^A.(\underline{\Gamma, x : A \vdash B})$$

$$(SFD) \quad \underline{\Gamma, x : A \vdash p} \to \mathsf{app}(x, \underline{\Gamma, x : A \rhd A}, p)$$

$$(KFD) \quad \mathsf{app}(H, \underline{\Gamma \rhd A \supset B}, p) \to \mathsf{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \rhd B}, p)$$

$$(FFD) \quad \mathsf{app}(H, \underline{\Gamma \rhd p}, p) \to \mathsf{app}(H)$$

Comments

NJT_∂: Reduction Rules

$$
\begin{array}{lr}
(IR) & \underline{\Gamma \vdash A \supset B} \to \lambda x^A.(\underline{\Gamma, x : A \vdash B}) \\
(SFD) & \underline{\Gamma, x : A \vdash p} \to \mathsf{app}(x, \underline{\Gamma, x : A \rhd A}, p) \\
(KFD) & \mathsf{app}(H, \underline{\Gamma \rhd A \supset B}, p) \to \mathsf{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \rhd B}, p) \\
(FFD) & \mathsf{app}(H, \underline{\Gamma \rhd p}, p) \to \mathsf{app}(H)
\end{array}
$$

Comments

- **One reduction rule for each inference rule;**

NJT_∂: Reduction Rules

$$(IR) \qquad \underline{\Gamma \vdash A \supset B} \to \lambda x^A.(\underline{\Gamma, x : A \vdash B})$$

$$(SFD) \qquad \underline{\Gamma, x : A \vdash p} \to \mathsf{app}(x, \underline{\Gamma, x : A \rhd A}, p)$$

$$(KFD) \qquad \mathsf{app}(H, \underline{\Gamma \rhd A \supset B}, p) \to \mathsf{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \rhd B}, p)$$

$$(FFD) \qquad \mathsf{app}(H, \underline{\Gamma \rhd p}, p) \to \mathsf{app}(H)$$

Comments

- One reduction rule for each inference rule;
- **Reduction rules tell how to use the inference rules during proof search;**

NJT_∂: Reduction Rules

$$(IR) \quad \underline{\Gamma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B})$$

$$(SFD) \quad \underline{\Gamma, x : A \vdash p} \rightarrow \mathsf{app}(x, \underline{\Gamma, x : A \rhd A}, p)$$

$$(KFD) \quad \mathsf{app}(H, \underline{\Gamma \rhd A \supset B}, p) \rightarrow \mathsf{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \rhd B}, p)$$

$$(FFD) \quad \mathsf{app}(H, \underline{\Gamma \rhd p}, p) \rightarrow \mathsf{app}(H)$$

Comments

- One reduction rule for each inference rule;
- Reduction rules tell how to use the inference rules during proof search;
- **Three distinct concepts: redex, formal sequent and proof obligation;**

NJT_∂: Reduction Rules

$$(IR) \qquad \underline{\Gamma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B})$$

$$(SFD) \qquad \underline{\Gamma, x : A \vdash p} \rightarrow \mathsf{app}(x, \underline{\Gamma, x : A \rhd A}, p)$$

$$(KFD) \qquad \mathsf{app}(H, \underline{\Gamma \rhd A \supset B}, p) \rightarrow \mathsf{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \rhd B}, p)$$

$$(FFD) \qquad \mathsf{app}(H, \underline{\Gamma \rhd p}, p) \rightarrow \mathsf{app}(H)$$

Comments

- One reduction rule for each inference rule;
- Reduction rules tell how to use the inference rules during proof search;
- Three distinct concepts: redex, formal sequent and proof obligation;
- **Reduction builds the record of the search.**

Example

$$\dfrac{}{\vdash (p \supset p) \supset p \supset p}$$

$$\vdash (p \supset p) \supset p \supset p$$

Example

$$\rightarrow_{IR} \qquad \frac{\vdash (p \supset p) \supset p \supset p}{\lambda f^{(p \supset p)}.(\underline{f : (p \supset p) \vdash p \supset p})}$$

$$\frac{f : p \supset p \vdash p \supset p}{\vdash (p \supset p) \supset p \supset p}\ I$$

Example

$$
\begin{array}{ll}
 & \underline{\vdash (p \supset p) \supset p \supset p} \\
\to_{IR} & \lambda f^{(p \supset p)}.(\underline{f : (p \supset p) \vdash p \supset p}) \\
\to_{IR} & \lambda f^{(p \supset p)}.\lambda x^{p}.(\underline{f : (p \supset p), x : p \vdash p})
\end{array}
$$

$$
\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\ I}{\vdash (p \supset p) \supset p \supset p}\ I
$$

Example

$$\underline{\vdash (p \supset p) \supset p \supset p}$$

$$\rightarrow_{IR} \quad \lambda f^{(p \supset p)}.(\underline{f : (p \supset p) \vdash p \supset p})$$

$$\rightarrow_{IR} \quad \lambda f^{(p \supset p)}.\lambda x^{p}.(\underline{f : (p \supset p), x : p \vdash p})$$

$$\rightarrow_{SFD} \quad \lambda f^{(p \supset p)}.\lambda x^{p}.\mathsf{app}(f, \underline{\Gamma \rhd p \supset p}, p)$$

$$\frac{}{\Gamma \rhd p \supset p}\ A$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\ I}{\vdash (p \supset p) \supset p \supset p}\ I$$

$$\Gamma = \{f : p \supset p, x : p\}$$

Example

$$\underline{\vdash (p \supset p) \supset p \supset p}$$

$$\to_{IR} \quad \lambda f^{(p\supset p)}.(\underline{f : (p \supset p) \vdash p \supset p})$$

$$\to_{IR} \quad \lambda f^{(p\supset p)}.\lambda x^{p}.(\underline{f : (p \supset p), x : p \vdash p})$$

$$\to_{SFD} \quad \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f, \underline{\Gamma \rhd p \supset p}, p)$$

$$\to_{KFD} \quad \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f(\underline{\Gamma \vdash p}), \underline{\Gamma \rhd p}, p)$$

$$\dfrac{\dfrac{}{\Gamma \rhd p \supset p}\,A \qquad \Gamma \vdash p}{\Gamma \rhd p}\,E$$

$$\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I$$

$$\Gamma = \{f : p \supset p, x : p\}$$

Example

$$
\begin{array}{ll}
 & \underline{\vdash (p \supset p) \supset p \supset p} \\
\rightarrow_{IR} & \lambda f^{(p \supset p)}.(\underline{f : (p \supset p) \vdash p \supset p}) \\
\rightarrow_{IR} & \lambda f^{(p \supset p)}.\lambda x^{p}.(\underline{f : (p \supset p), x : p \vdash p}) \\
\rightarrow_{SFD} & \lambda f^{(p \supset p)}.\lambda x^{p}.\mathsf{app}(f, \underline{\Gamma \rhd p \supset p}, p) \\
\rightarrow_{KFD} & \lambda f^{(p \supset p)}.\lambda x^{p}.\mathsf{app}(f(\underline{\Gamma \vdash p}), \underline{\Gamma \rhd p}, p)
\end{array}
$$

$$
\dfrac{\dfrac{}{\Gamma \rhd p \supset p}\,A \qquad \Gamma \vdash p}{f : p \supset p, x : p \rhd p}\,E
$$

$$
\dfrac{\dfrac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I
$$

$$\Gamma = \{f : p \supset p, x : p\}$$

Example

$$
\begin{array}{ll}
 & \underline{\vdash (p \supset p) \supset p \supset p} \\
\rightarrow_{IR} & \lambda f^{(p\supset p)}.(\underline{f : (p \supset p) \vdash p \supset p}) \\
\rightarrow_{IR} & \lambda f^{(p\supset p)}.\lambda x^{p}.(\underline{f : (p \supset p), x : p \vdash p}) \\
\rightarrow_{SFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f, \underline{\Gamma \rhd p \supset p}, p) \\
\rightarrow_{KFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f(\underline{\Gamma \vdash p}), \underline{\Gamma \rhd p}, p) \\
\rightarrow_{FFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f(\underline{\Gamma \vdash p}))
\end{array}
$$

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \rhd p \supset p}\,A \qquad \Gamma \vdash p}{f : p \supset p, x : p \rhd p}\,E}{f : p \supset p, x : p \vdash p}\,C}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I
$$

$$\Gamma = \{f : p \supset p, x : p\}$$

Example

$$
\begin{array}{ll}
 & \underline{\vdash (p \supset p) \supset p \supset p} \\
\to_{IR} & \lambda f^{(p\supset p)}.(\underline{f : (p \supset p) \vdash p \supset p}) \\
\to_{IR} & \lambda f^{(p\supset p)}.\lambda x^{p}.(\underline{f : (p \supset p), x : p \vdash p}) \\
\to_{SFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f, \underline{\Gamma \rhd p \supset p}, p) \\
\to_{KFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f(\underline{\Gamma \vdash p}), \underline{\Gamma \rhd p}, p) \\
\to_{FFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f(\underline{\Gamma \vdash p})) \\
\to_{SFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f\mathsf{app}(x, \underline{\Gamma \rhd p}, p)) \\
\to_{FFD} & \lambda f^{(p\supset p)}.\lambda x^{p}.\mathsf{app}(f\mathsf{app}(x))
\end{array}
$$

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \rhd p \supset p}\,A \qquad \dfrac{\dfrac{}{\Gamma \rhd p}\,A}{\Gamma \vdash p}\,C}{f : p \supset p, x : p \rhd p}\,E}{f : p \supset p, x : p \vdash p}\,C}{f : p \supset p \vdash p \supset p}\,I}{\vdash (p \supset p) \supset p \supset p}\,I
$$

$$\Gamma = \{f : p \supset p, x : p\}$$

Results

Results

Conservativity

NJT derives $\Gamma \vdash M : A$ **iff** NJT_∂ derives $\epsilon \Vdash M : (\Gamma \vdash A)$.

Results

Conservativity

$\boxed{\textit{NJT} \text{ derives } \Gamma \vdash M : A}$ **iff** $\textit{NJT}_\partial$ derives $\epsilon \Vdash M : (\Gamma \vdash A)$.

Results

Conservativity

$\boxed{\textit{NJT} \text{ derives } \Gamma \vdash M : A}$ **iff** $\boxed{\textit{NJT}_\partial \text{ derives } \epsilon \Vdash M : (\Gamma \vdash A)}$.

Results

Conservativity

$\boxed{\textit{NJT} \text{ derives } \Gamma \vdash M : A}$ **iff** $\boxed{\textit{NJT}_\partial \text{ derives } \epsilon \Vdash M : (\Gamma \vdash A)}$.

Results

Conservativity

NJT derives $\Gamma \vdash M : A$ **iff** *NJT*$_\partial$ derives $\epsilon \Vdash M : (\Gamma \vdash A)$.

Record of Search

If $\Xi \Vdash M : \sigma$ **then** $\underline{\sigma} \twoheadrightarrow M$.

Results

Conservativity

NJT derives $\Gamma \vdash M : A$ **iff** *NJT*$_\partial$ derives $\epsilon \Vdash M : (\Gamma \vdash A)$.

Record of Search

If $\Xi \Vdash M : \sigma$ **then** $\underline{\sigma} \twoheadrightarrow M$.

Subject Reduction

If $\Xi \Vdash M : \sigma$ and $M \rightarrow M'$ **then** there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash M' : \sigma$.

Results

Results

Proof Search as Normalization

NJT derives $\Gamma \vdash M : A$ **iff** $\underline{\Gamma \vdash A} \twoheadrightarrow M$.

Unrestricted Intercalation in Natural Deduction

Recall *NJT*

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B}\ I \qquad \frac{\Gamma \rhd p}{\Gamma \vdash p}\ C$$

$$\frac{}{\Gamma, x : A \rhd A}\ A \qquad \frac{\Gamma \rhd A \supset B \quad \Gamma \vdash A}{\Gamma \rhd B}\ E$$

Intercalation Calculus

Mix Inversion and Focusing Phases

Intercalation Calculus

Mix Inversion and Focusing Phases

$$\Gamma \vdash q \supset r$$

$$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$$

Intercalation Calculus

Mix Inversion and Focusing Phases

$$\dfrac{\dfrac{}{\Gamma \rhd p \supset (q \supset r)}\,A \qquad \overset{?}{\Gamma \vdash p}}{\Gamma \rhd q \supset r}\,E$$

$$\Gamma \vdash q \supset r$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Mix Inversion and Focusing Phases

?

$$\dfrac{\dfrac{}{\Gamma \rhd p \supset (q \supset r)}\, A \qquad \Gamma \vdash p}{\Gamma \rhd q \supset r}\, E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\, I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Mix Inversion and Focusing Phases

?

$$\dfrac{\dfrac{}{\Gamma \rhd p \supset (q \supset r)}\,A \qquad \Gamma \vdash p}{\Gamma \rhd q \supset r}\,E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\,I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Mix Inversion and Focusing Phases

?

$$\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\, A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \rhd q \supset r}\, E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\, I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Mix Inversion and Focusing Phases

?

?

$$\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \rhd q \supset r}\,E \qquad \Gamma, x : q \vdash q}{\Gamma, x : q \rhd r}\,E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\,I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Mix Inversion and Focusing Phases

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \rhd q \supset r}\,E \qquad \Gamma, x : q \vdash q}{\Gamma, x : q \rhd r}\,E}{\Gamma, x : q \vdash r}\,C}{\Gamma \vdash q \supset r}\,I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Mix Inversion and Focusing Phases

$$
\dfrac{
  \dfrac{
    \dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A
    \qquad
    \dfrac{\dfrac{}{\Gamma, x : q \rhd p}\,A}{\Gamma, x : q \vdash p}\,C
  }{\Gamma, x : q \rhd q \supset r}\,E
  \qquad
  \dfrac{\dfrac{}{\Gamma, x : q \rhd q}\,A}{\Gamma, x : q \vdash q}\,C
}{
  \dfrac{
    \dfrac{\Gamma, x : q \rhd r}{\Gamma, x : q \vdash r}\,C
  }{\Gamma \vdash q \supset r}\,I
}\,E
$$

$$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$$

Intercalation Calculus

Parallel Focusing

Intercalation Calculus

Parallel Focusing

$$\Gamma \vdash q \supset r$$

$$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$$

Intercalation Calculus

Parallel
Focusing

$$\frac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\ I$$

$$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$$

Intercalation Calculus

Parallel Focusing

?

$$\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\ A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \rhd q \supset r}\ E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\ I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Parallel Focusing

$$\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\ A \qquad \overset{?}{\Gamma, x : q \vdash p}}{\Gamma, x : q \rhd q \supset r}\ E$$

$$\dfrac{\dfrac{}{\Gamma, x : q \rhd q \supset s}\ A \qquad \overset{?}{\Gamma, x : q \vdash q}}{\Gamma, x : q \rhd s}\ E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\ I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Parallel Focusing

$$
\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \rhd q \supset r}\,E \qquad \Gamma, x : q \vdash q}{\Gamma, x : q \rhd r}\,E
\qquad
\dfrac{\dfrac{}{\Gamma, x : q \rhd q \supset s}\,A \qquad \Gamma, x : q \vdash q}{\Gamma, x : q \rhd s}\,E
$$

$$
\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\,I
$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Parallel Focusing

$$\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \triangleright p \supset (q \supset r)}\,A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \triangleright q \supset r}\,E \qquad \Gamma, x : q \vdash q}{\Gamma, x : q \triangleright r}\,E \qquad \dfrac{\dfrac{}{\Gamma, x : q \triangleright q \supset s}\,A \qquad \Gamma, x : q \vdash q}{\Gamma, x : q \triangleright s}\,E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\,I$$

$$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$$

Intercalation Calculus

Parallel Focusing

$$\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A \qquad \overset{?}{\Gamma, x : q \vdash p}}{\Gamma, x : q \rhd q \supset r}\,E \qquad \overset{?}{\Gamma, x : q \vdash q}}{\Gamma, x : q \rhd r}\,E$$

$$\dfrac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r}\,I$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Parallel Focusing

$$
\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A \qquad \Gamma, x : q \vdash p}{\Gamma, x : q \rhd q \supset r}\,E \qquad \Gamma, x : q \vdash q}{\dfrac{\Gamma, x : q \rhd r}{\Gamma, x : q \vdash r}\,C}\,E}{\Gamma \vdash q \supset r}\,I
$$

$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$

Intercalation Calculus

Parallel Focusing

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma, x : q \rhd p \supset (q \supset r)}\,A \qquad \dfrac{\dfrac{}{\Gamma, x : q \rhd p}\,A}{\Gamma, x : q \vdash p}\,C}{\Gamma, x : q \rhd q \supset r}\,E \qquad \dfrac{\dfrac{}{\Gamma, x : q \rhd q}\,A}{\Gamma, x : q \vdash q}\,C}{\Gamma, x : q \rhd r}\,E}{\Gamma, x : q \vdash r}\,C}{\Gamma \vdash q \supset r}\,I
$$

$$\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$$

IC

Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H)$$
$$H ::= x \mid H N$$

IC

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H) \mid \underline{\Sigma \vdash A}$$
$$H ::= x \mid H N$$

IC

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H) \mid \underline{\Sigma \vdash A}$$
$$H ::= x \mid H N$$

Saturation Set

$$\Sigma ::= \emptyset \mid \Sigma \cup \{(H : B)\}$$

IC

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H) \mid \underline{\Sigma \vdash A}$$

$$H ::= x \mid H\,N$$

Saturation Set

$$\Sigma ::= \emptyset \mid \Sigma \cup \{(\underline{H} : B)\}$$

IC

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H) \mid \underline{\Sigma \vdash A}$$
$$H ::= x \mid H N$$

Generalization of NJT_∂

$$\underline{\sigma} = \underline{\Gamma \vdash A}$$

IC

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H) \mid \underline{\Sigma \vdash A}$$
$$H ::= x \mid H N$$

Generalization of NJT_∂

$$\underline{\sigma} = \underline{\Gamma \vdash A}$$
$$app(H, \underline{\Gamma \rhd A}, p) = \underline{\Gamma, H : A \vdash p}$$

IC

Partial Proof Terms

$$M, N ::= \lambda x^A.M \mid app(H) \mid \underline{\Sigma \vdash A}$$
$$H ::= x \mid H N$$

Partial Sequents

$$\Vdash M : (\Gamma \vdash A)$$
$$\Vdash \Sigma : \Gamma$$

IC

Inference Rules

$$\frac{\Vdash M : (\Gamma, x : A \vdash B)}{\Vdash \lambda x^A . M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Vdash \Sigma : \Gamma \quad (H : p) \in \Sigma}{\Vdash app(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\Vdash \Gamma : \Gamma}\ A \qquad \frac{\Vdash (\Sigma, H : A \supset B) : \Gamma \quad \Vdash N : (\Gamma \vdash A)}{\Vdash (\Sigma, HN : B) : \Gamma}\ E*$$

* if $H = x$, then $(H : A \supset B) \in \Sigma$

IC

Inference Rules

$$\frac{\Vdash \Sigma : \Gamma}{\Vdash \underline{\Sigma \vdash A} : (\Gamma \vdash A)}\ \partial$$

$$\frac{\Vdash M : (\Gamma, x : A \vdash B)}{\Vdash \lambda x^{A}.M : (\Gamma \vdash A \supset B)}\ I \qquad \frac{\Vdash \Sigma : \Gamma \quad (H : p) \in \Sigma}{\Vdash app(H) : (\Gamma \vdash p)}\ C$$

$$\frac{}{\Vdash \Gamma : \Gamma}\ A \qquad \frac{\Vdash (\Sigma, H : A \supset B) : \Gamma \quad \Vdash N : (\Gamma \vdash A)}{\Vdash (\Sigma, HN : B) : \Gamma}\ E^{*}$$

* if $H = x$, then $(H : A \supset B) \in \Sigma$

IC: Reduction Rules

$$(IR) \qquad \underline{\Sigma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B})$$

$$(S)\text{*} \qquad \Sigma, H : A \supset B \rightarrow \Sigma, H(\underline{\Gamma \vdash A}) : B \quad , \Sigma \sqsupseteq \Gamma$$

$$(FS) \qquad \underline{\Sigma, H : p \vdash p} \rightarrow app(H)$$

* if $H = x$, then $(H : A \supset B) \in \Sigma$

IC: Reduction Rules

$$(IR) \qquad \underline{\Sigma \vdash A \supset B} \to \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B})$$

$$(S)^* \qquad \Sigma, H : A \supset B \to \Sigma, H(\underline{\Gamma \vdash A}) : B \quad , \Sigma \sqsupseteq \Gamma$$

$$(FS) \qquad \underline{\Sigma, H : p \vdash p} \to app(H)$$

Comments

- **$\Sigma \sqsupseteq \Gamma$ means that Γ is determined from Σ;**

* if $H = x$, then $(H : A \supset B) \in \Sigma$

IC: Reduction Rules

$$(IR) \qquad \underline{\Sigma \vdash A \supset B} \to \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B})$$

$$(S)^* \qquad \Sigma, H : A \supset B \to \Sigma, H(\underline{\Gamma \vdash A}) : B \quad , \Sigma \sqsupseteq \Gamma$$

$$(FS) \qquad \underline{\Sigma, H : p \vdash p} \to app(H)$$

Comments

- $\Sigma \sqsupseteq \Gamma$ means that Γ is determined from Σ;
- **$\omega_{x:A}\,(\Sigma)$ does a global update that is not carried in Sieg's IC.**

* if $H = x$, then $(H : A \supset B) \in \Sigma$

IC: Reduction Rules

$$(IR) \qquad \underline{\Sigma \vdash A \supset B} \to \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B})$$

$$(S)^* \qquad \Sigma, H : A \supset B \to \Sigma, H(\underline{\Gamma \vdash A}) : B \quad , \Sigma \sqsupseteq \Gamma$$

$$(FS) \qquad \underline{\Sigma, H : p \vdash p} \to app(H)$$

Comments

- $\Sigma \sqsupseteq \Gamma$ means that Γ is determined from Σ;
- $\omega_{x:A}(\Sigma)$ does a global update that is not carried in Sieg's IC.
- $\Sigma, H : A \supset B \to \Sigma, H(\underline{\Sigma, H : A \supset B \vdash A}) : B$ * is derivable.

* if $H = x$, then $(H : A \supset B) \in \Sigma$

Results

Conservativity

NJT derives $\Gamma \vdash M : A$ **iff** IC derives $\Vdash M : (\Gamma \vdash A)$.

Record of Search

If $\Vdash M : (\Gamma \vdash A)$ **then** $\underline{\Gamma \vdash A} \twoheadrightarrow M$.

Subject Reduction

If $\Vdash M : (\Gamma \vdash A)$ and $M \to M'$ **then** $\Vdash M' : (\Gamma \vdash A)$.

Results

Proof Search as Normalization

NJT derives $\Gamma \vdash M : A$ **iff** IC derives $\underline{\Gamma \vdash A} \twoheadrightarrow M$.

Conclusions and Future Work

Contributions

- **Update of Sieg's Intercalation Calculus:**

Contributions

- Update of Sieg's Intercalation Calculus:
    - **integration with bidirectional natural deduction;**

Contributions

- Update of Sieg's Intercalation Calculus:
  - integration with bidirectional natural deduction;
  - **identification of focused fragments (eg: NJT_∂);**

Contributions

- Update of Sieg's Intercalation Calculus:
  - integration with bidirectional natural deduction;
  - identification of focused fragments (eg: NJT_∂);
  - **formalization of proof-search rules as rewriting rules on PPTs.**

Future Work

- IC:
    - **extension to classical logic;**

Future Work

- IC:
    - extension to classical logic;
    - **correspondence with a sequent calculus system;**

Future Work

- IC:
  - extension to classical logic;
  - correspondence with a sequent calculus system;
  - **possible generalization of focusing and intercalation equivalence;**

Future Work

- IC:
  - extension to classical logic;
  - correspondence with a sequent calculus system;
  - possible generalization of focusing and intercalation equivalence;
  - **additional subsystems and fragments.**

Thank you!

Obrigada!

An Intercalation Calculus with Partial Proof Terms

Ana Catarina Sousa
LIACC - Artificial Intelligence and Computer Science Laboratory

Joint work with José Espírito Santo

Women in Logic Workshop
Lisbon, Portugal

24th July 2026